

Arunabh Bhattacharyya

Data Scientist & Full-Stack Developer | ML · Computer Vision · RAG · IoT

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EDUCATION

Dibrugarh University Institute of Engineering & Technology (DUIET)

Aug 2023 – Present

B.Tech – Computer Science & Engineering | Expected 2027

Relevant: DSA, Operating Systems, Computer Networks, DBMS, Compiler Design, Theory of Computation

INTERSHIP EXPERIENCE

Summer Research Intern — Technology Innovation Hub, IIT Guwahati (TI&DF)

Jun – Jul 2025

Cert. No. IITG/TIDF/SI2025/037

- Built end-to-end computer vision pipeline for real-time underwater trash detection using Python, OpenCV, TensorFlow/Keras, and YOLOv5, achieving >88% detection precision on curated test set.
- Deployed interactive Streamlit web app for live inference; demo published on LinkedIn and validated by IIT Guwahati faculty.
- Collaborated directly with Prof. S.K. Dwivedy (Project Director) and Mr. N.N. Dutta (CEO, TI&DF), gaining exposure to applied ML research workflows.

COMPETITIONS & AWARDS

[TOP] TOP Team — GUenARK 2026 SIH 1.0 Hackathon

Prize: Rs. 50,500 | Feb 2026

Team: Buri Buri Zaemon | Gauhati University | Sponsors: Adani Power, Mahindra, NRL, Tata Motors

- Awarded TOP Team at 24-hour hackathon by GU Innovation, Incubation & Startup Cell; recognised for high-complexity prototype execution; certificate signed by Vice Chancellor.

[1st] Winner — "Algo Arena" Hackathon

Prize: Rs. 12,000 | Nov 2025

Team: Buri Buri Jai Mon | CTII, Dibrugarh University (19–21 Nov 2025)

- Secured 1st position; certificate signed by Faculty Coordinator and Founder & CEO, DSRL.

[Finalist] 4th Position — All Assam Hackathon

Dibrugarh University

[4th] 4th Position — CRITICAL MASS: Ultimate AI Bot Making Competition

Mar 2026

Team: LazyCoders | Apeiron Fest, NJACK — IIT Patna | Online

- Engineered a Chain Reaction game-playing AI bot using Minimax search with Alpha-Beta Pruning and Iterative Deepening, achieving 3–5 ply lookahead and evaluating up to 100,000 nodes/move within a strict 1-second constraint; pruned ~98.5% of the search space.

PROJECTS

CHAI-NET — Intelligent Tea Garden Management System

Feb 2026

FastAPI · Next.js 14 · TensorFlow · YOLOv5 · Gemini AI · IoT · PostgreSQL

- Engineered AI-powered IoT platform for Assam tea estates; integrated real-time sensor dashboard, CV-based leaf disease scanner, and ML market price forecasting, reducing crop loss risk by an estimated 30%.
- Implemented Farmer Action Simulator enabling data-driven cultivation decisions; shipped full-stack product with Next.js 14 frontend and FastAPI backend. github.com/EnderMRG/Tea

BankAssist AI — Private LLM Assistant for Banks

Mar 2026

RAG · Python · Ollama · LangChain · LLM · PDF/DOCX/XLSX Ingestion

- Built 100% air-gapped AI assistant for bank staff grounded in RBI/NABARD/SEBI circulars — zero data leaves the bank's network, ensuring full regulatory compliance.
- RAG architecture with precise source citation (Circular Ref No., Date, Issuing Authority); supports ingestion of PDF, DOCX, XLSX; cut policy lookup time by ~60% in internal testing. github.com/runabh1/bank-llm

AI-Powered Downhole Equipment Anomaly Detector

Dec 2025

Python · Isolation Forest · Pandas · NumPy · Streamlit

- Designed unsupervised ML pipeline (Isolation Forest) for anomaly detection on oil & gas downhole sensor data, flagging critical failure patterns with >90% recall on validation set.

- Built real-time Streamlit dashboard for anomaly visualization; proactive alerts reduced simulated unplanned downtime by ~35%.
github.com/runabh1/downhole-anomaly-equipment-detector

TECHNICAL SKILLS

Languages	Python, C, C++, TypeScript, JavaScript, SQL, HTML, CSS
AI / ML	TensorFlow, Keras, Scikit-learn, OpenCV, YOLOv5/v8, Isolation Forest, Random Forest, RAG, Gemini API, Pandas, NumPy, Matplotlib
Web & Backend	FastAPI, Next.js 14, React, Node.js, Express.js, PostgreSQL, Prisma, MongoDB, Streamlit, Firebase
Tools	Git, GitHub, Linux/Unix, VS Code, Tailwind CSS, LaTeX, Ollama, LangChain
Core CS	Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Compiler Design, Theory of Computation

CERTIFICATIONS & TRAINING

- Full Stack Web Development Bootcamp — Udemy (Dr. Angela Yu)**

Aug 2025 – Present
- Covers React, Next.js, Node.js, Express, PostgreSQL via project-based curriculum.

EXTRA-CURRICULAR ACTIVITIES

- Design Team Member — Google Developer Student Clubs (GDSC), Dibrugarh University**

Present
- Lead visual identity for community events (500+ members); produce graphics, posters, and digital content in collaboration with dev and content teams.